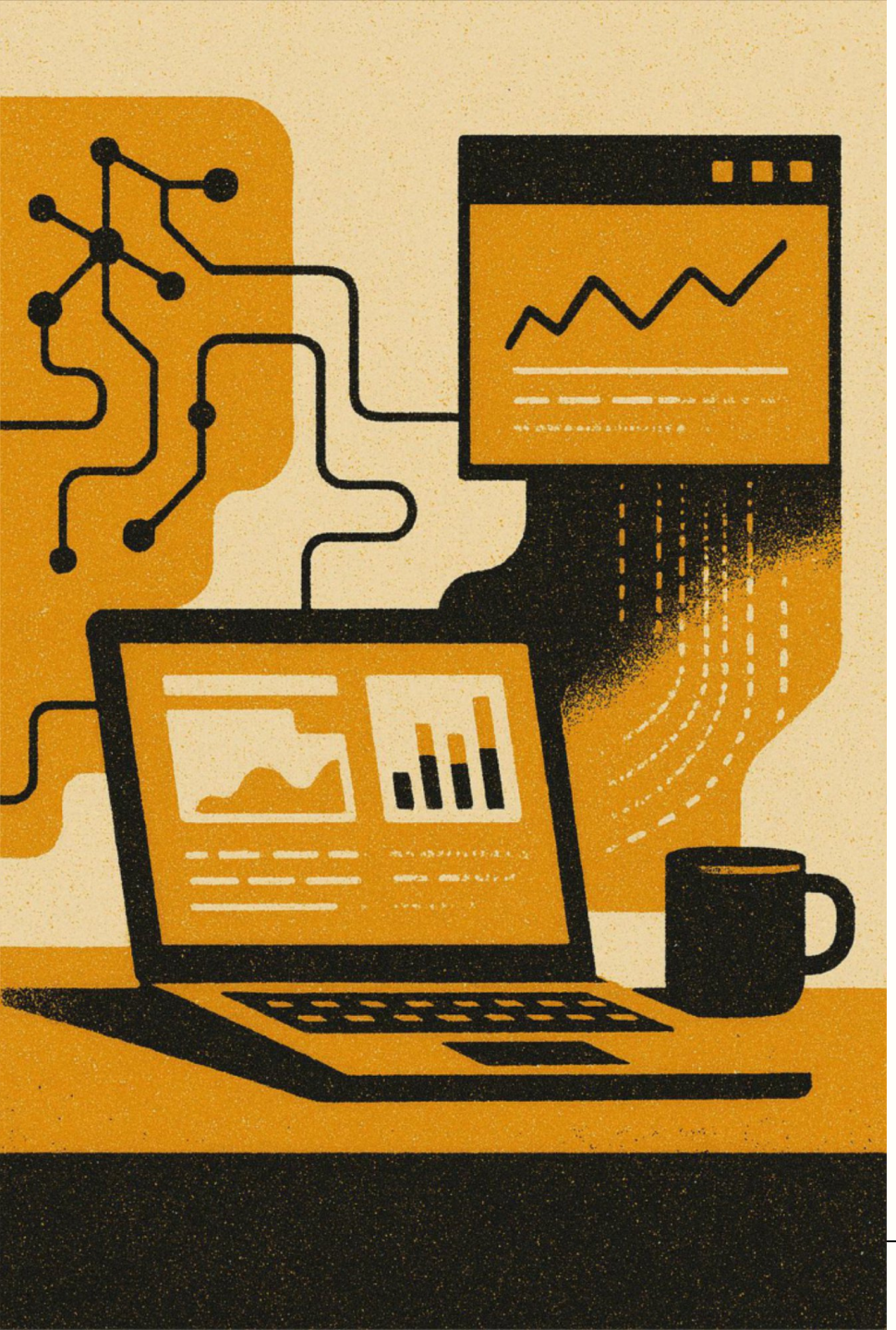
Building Agentic AI on ELK Stack

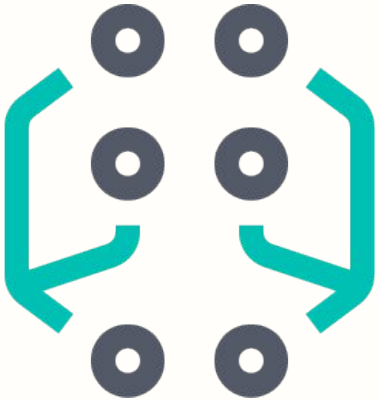


Building Agentic AI on ELK Stack

Agenda

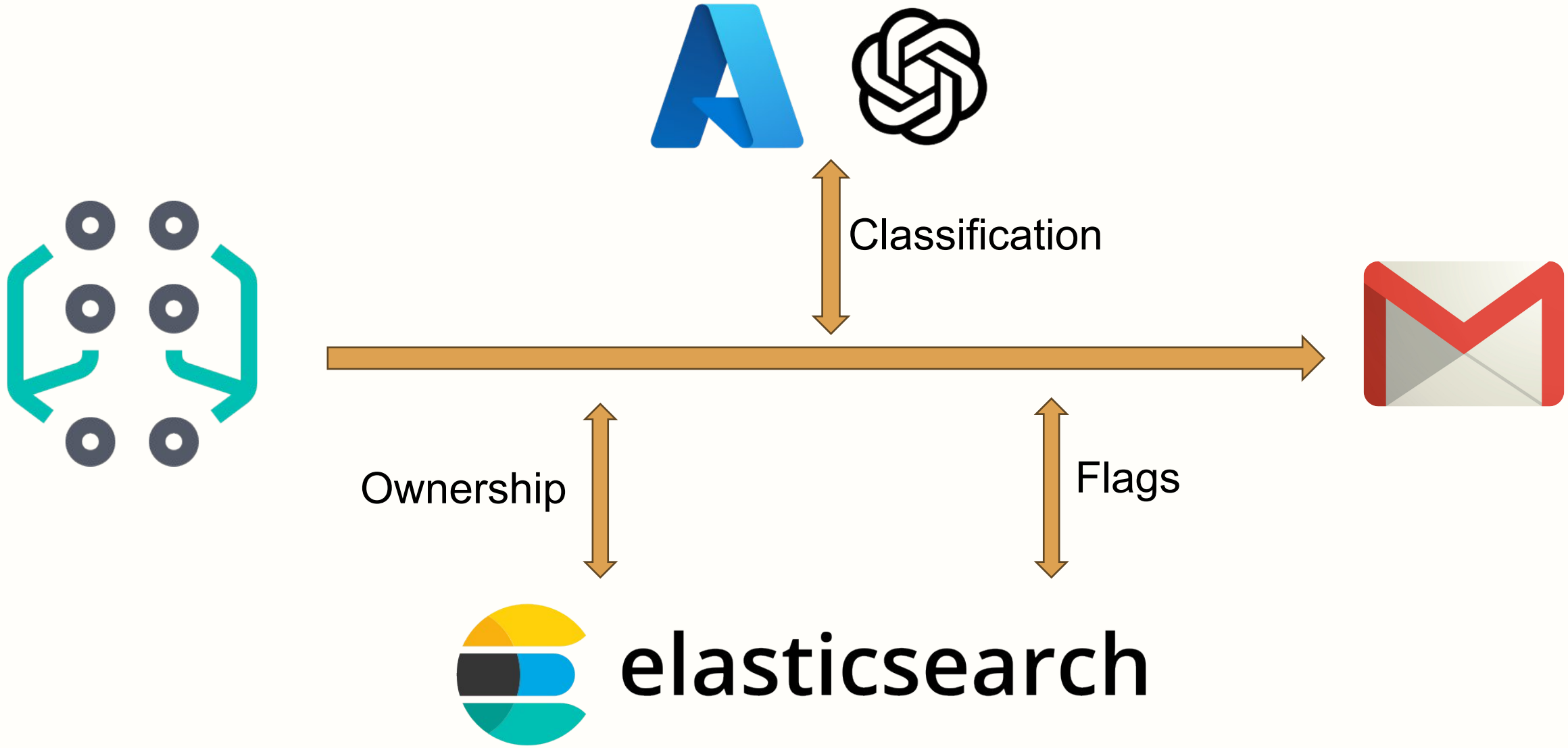
- Understanding ELK Stack and AI Agent
- Developing a Basic AI Agent
- Building Advanced AI Agents with Ruby
- Integrating AI Agents with Elastic UI
- Q & A

Anomaly notifications

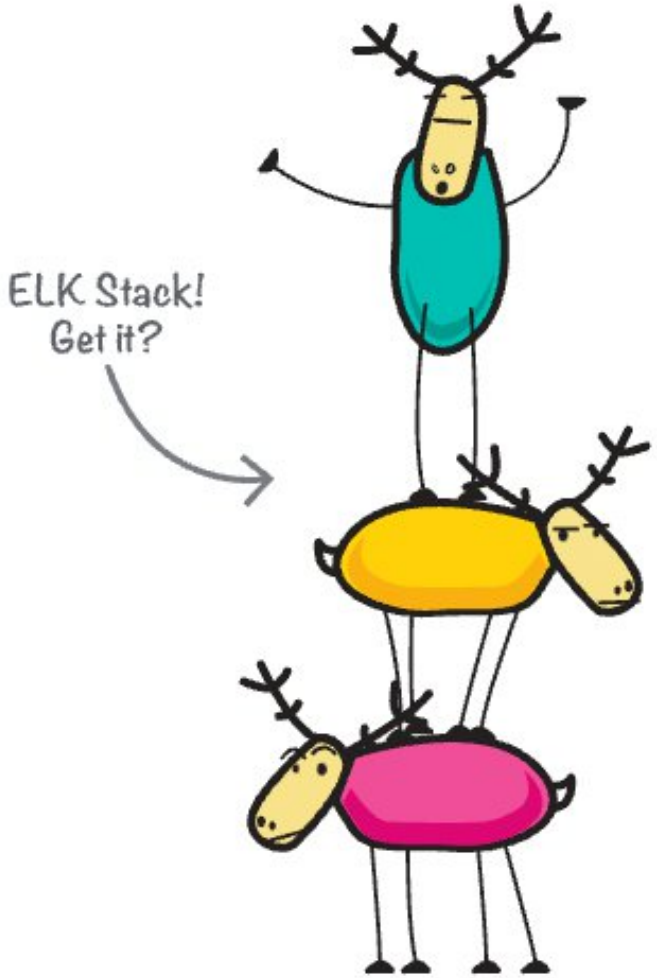


elasticsearch

Why AI Agent



Pure ELK Motivation

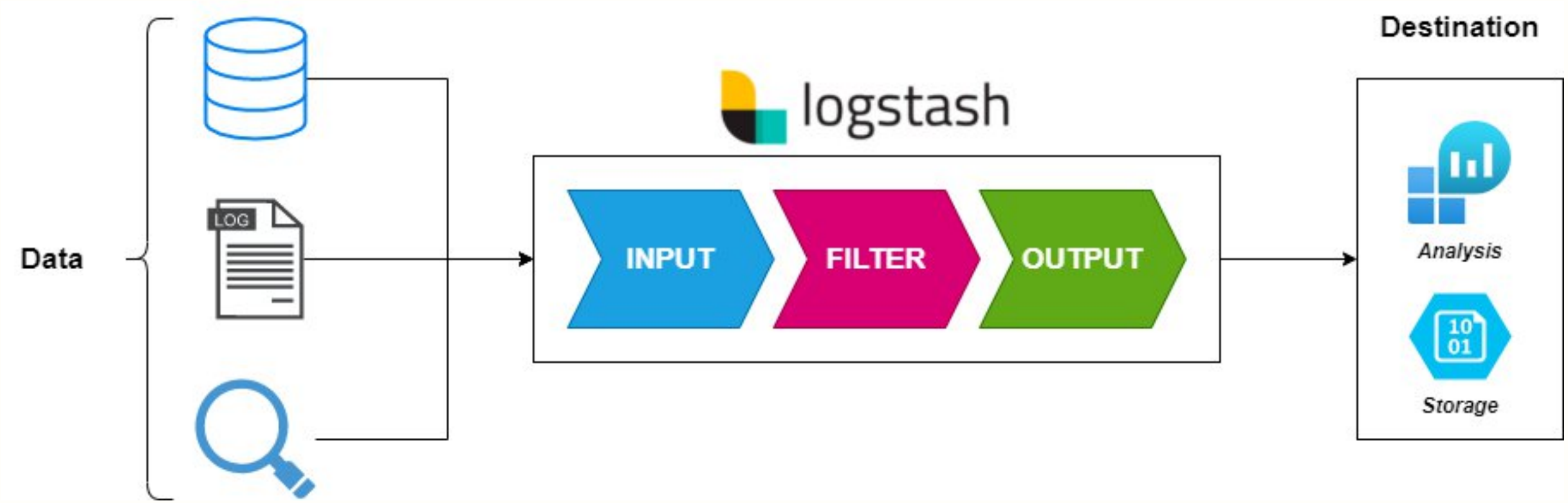


E Elasticsearch

L Logstash

K Kibana

What is Logstash



“Deliver at least once”

Logstash pipeline secrets

Input

logstash

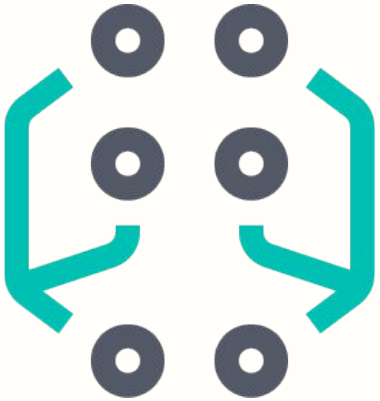
Filter

split
aggregate
drop
dissect
grok
json
elasticsearch
http
geoip
....

Output

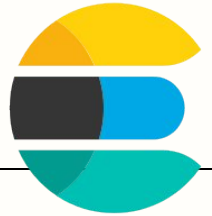
logstash

Logstash AI Agent - start



Anomaly

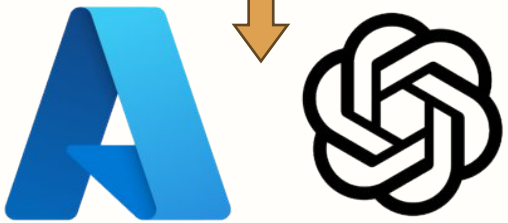
```
input {
        elasticsearch {
                schedule => "*/5 * * * *"
                query => '....
        }
}

filter {
        mutate {
                add_field => { "tags" => "exported" }
        }
}

output {
        pipeline {send_to => "processing_pipeline"}
        # write back to Elasticsearch
        elasticsearch {
            ...
        }
}
```



elasticsearch

Logstash AI Agent - processing

```
input {  pipeline {address => processing_pipeline} }

filter {
        # read full error log
         elasticsearch { … }

        # read Feature Flags around the anomaly time
         elasticsearch { … }

        # ask LLM to categorize error
         http {…}

        # find owner by category
         translate { … }
}
output { # Notify owner about anomaly
        email { … }
}
```



Logstash AI Agent – querying Elastic

```
# read  full error message around the anomaly time
elasticsearch {
        hosts => ["${ELASTIC_HOST}"]
        user => "${ELASTIC_USERNAME}"
        password => "${ELASTIC_PASSWORD}"
        index => "logs-core-clinical-*"
        query_template =>
"/usr/share/logstash/find_ms.json"
        fields => {
                "message" => "[alert][example]"
        }
}
if [alert][example] {
        # full error message sample found
        ….
}
```

elasticsearch

/usr/share/logstash/find_ms.json

```
{"size": 1,
"query": {"bool": {"must": [
    {"match_phrase": {
            "message": "%{[alert][terms]}"
        }
    },
    {
        "range": {
            "@timestamp": {
                "format": "strict_date_optional_time",
                "gte": "%{[timestamp_str]}-1h/h",
                "lte": "%{[timestamp_to_str]}+1h/h"
            }
        }
    }
]}}}
```

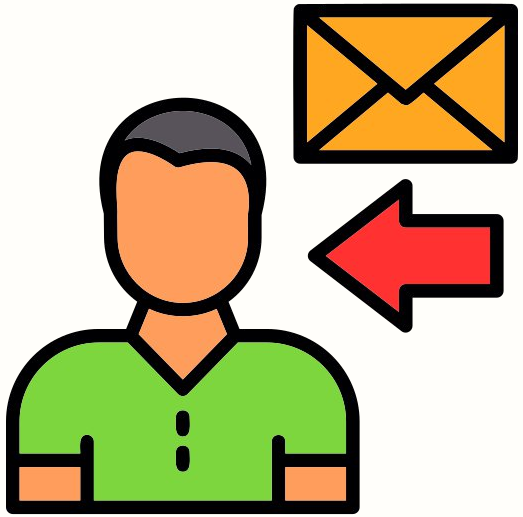
Logstash AI Agent – ask LLM

```
mutate {
  replace => {
    "[prompt]" =>
        "You are software engineer and need
to classify each application error to one of
Following categories: …"
  }
  ….
}
```

```
# ask LLM to categorize error
http {
        url => "${AZURE_OPENAI_ENDPOINT}"
        body_format => json
        verb => "POST"
        headers => {
        Accept => "application/json"
                "Content-Type" => "application/json"
                "api-key" => "${AZURE_OPENAI_KEY}"
        }
        body => '{"messages": [
                {"role": "system", "content":
            "%{prompt}"},
                {"role": "user",
                        "content": "%{[alert][example]}"}
                ]}
        '

        target_body => "[llm_response]"
}
```

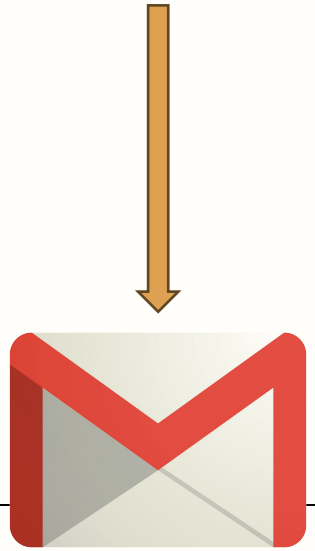
Logstash AI Agent - translation

```
if [llm_response][choices][0][message][content] {
    mutate {
      replace => {
        "[llm_ category]" => "%{[llm_response][choices][0][message][content]}"
      }
      lowercase => ["[llm_ category]"]
    }
    if [llm_ category] =~ /infrastructure/ {
        mutate {
          replace => {"[category]" =>" infrastructure "}
        }
    } else if [llm_ category] =~ /security/ {
    ....
    # find owner by category
    translate {
        source     => "[category]"
        target     => "[recipient]"
        dictionary_path => "/usr/share/logstash/config/categories.yaml'"
        fallback => "all_core_service_teams@pointclickcare.com"
    }
}
```



Logstash AI Agent – sending notification

```
output {
        email {
                to => ‘%{recipient}’
                from => elk_monitor@ pointclickcare.com’
                subject => 'Alert - %{category}’
                body => ”Abnormal error(s) detected: %{[alert][example]}  "
                template_file => "/tmp/email_template.mustache"
                domain => 'mail.pointclickcare.com’
                port => 25
        }
}
```



Use Cases & Limitations

Quick Anomaly Detection

Detects unusual patterns in log data instantly, enabling rapid response to operational issues.

Hard to debug and scale

Logstash does not support debugging

Hard to implement complex logic in pipelines

Automated Precise Alerting

Triggers alerts for specific teams, improving incident awareness and response times.

Lack of Adaptive Learning

Agents cannot evolve based on new data, limiting their effectiveness in dynamic environments.

Advanced Logstash AI Agent

```
ruby {
        init => '@llm_category_template=File.open("/usr/share/logstash/prompt.txt").read'
        code => '
                ms = [{"role": "system", "content": @ llm_category_template}]
                ms << {"role": "user", "content": "Here is the
error:\n"+event.get("[alert][example]")}
                event.set("[llm_request][messages]", ms)
'}

 ruby {
         path => "/usr/share/logstash/ai_agent.rb"
 }
```

Advanced Logstash AI Agent - debugging

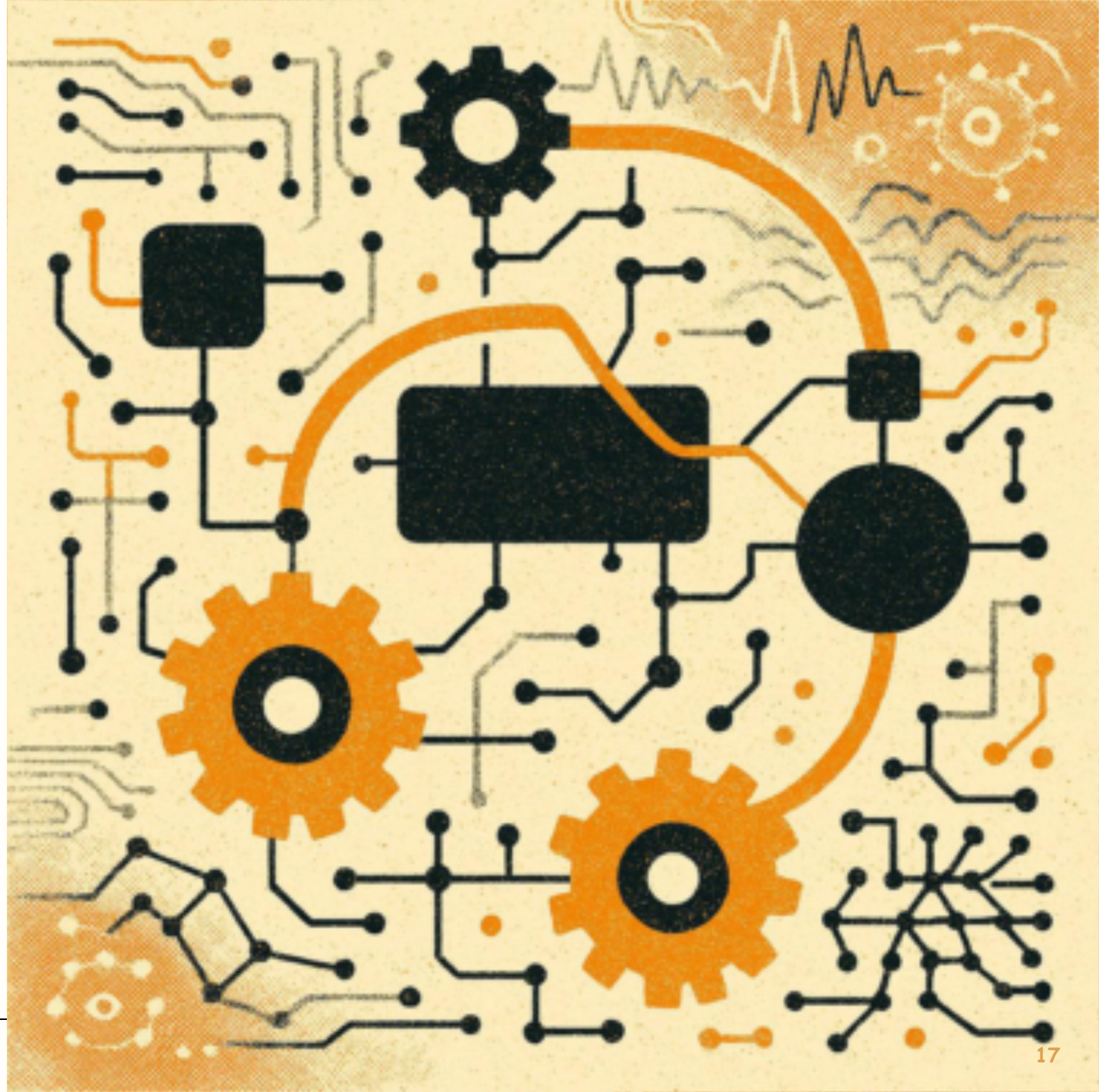
```
# code to debug Ruby AI Agent locally
require 'logstash-event'
require 'manticore'
require 'json'
require 'set'
require 'socket'
require 'uri'
require_relative 'ai_agent'

# initialize Plugin first
register(nil)

# create Event for processing
event = LogStash::Event.new
event.set("[alert][terms]", "SQL syntax
error")

# run Plugin once
filter(event)
```

ai_agent.rb

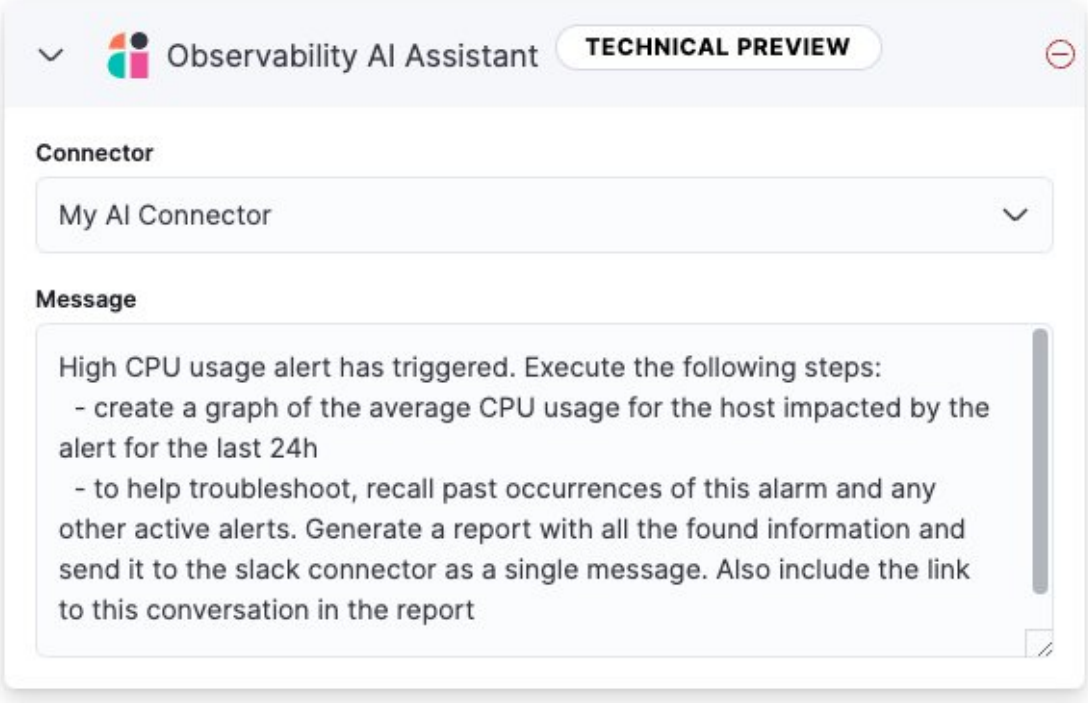
```
# logging methods to have consistent format
def log_error(message)
  puts ...
end

# This method is called when the plugin is loaded
def register(params)
        ...
end

# this method is called for each event
def filter(event)
        ...
        [event]
end
```

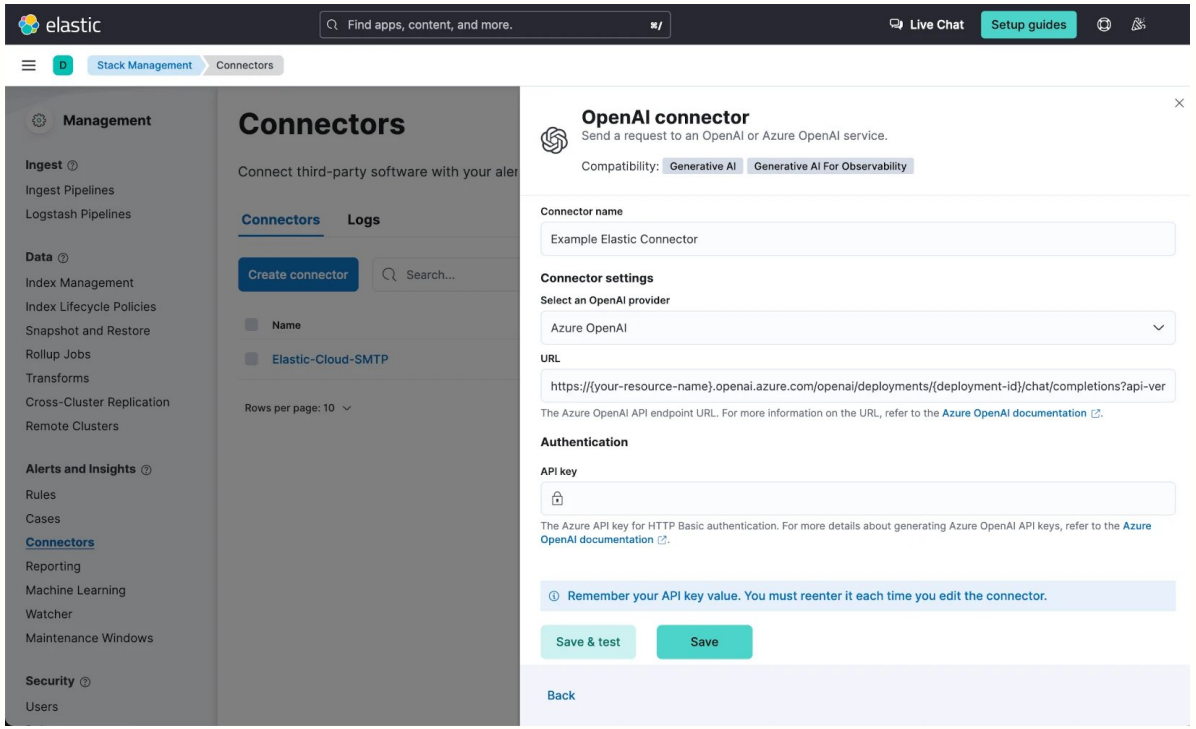
Logstash Ruby Plugin benefits

- Debugging + unit-testing
- Power of Ruby
- IDE with AI tools



Integrating AI Agent with Kibana





AI Agent as LLM

kibana

JRuby

```
def register(params)
  require "socket"
  require "uri"
  require "json"

 Thread.new do
     server = TCPServer.new ai_agent_port
        log_info("HTTP server on port
#{ai_agent_port}")
        loop do
            Thread.start(server.accept) do |socket|
begin
                request = socket.gets
                if request.nil?
                  socket.close
                  next
                end
                verb,path_and_query,protocol =
request.split(" ")
                uri_path = URI(path_and_query)
```

Unlocking AI Potential through ELK

Agentic AI with ELK

AI agents can be built effectively on the ELK stack within Elasticsearch environments.

Logstash + Ruby

Pipelines and the Ruby plugin enable intelligent behaviors and orchestration.

Elastic AI Assistant

Integration expands capabilities and accelerates practical applications.

Robust & Secure

A flexible approach to agentic AI aligned with Elasticsearch power and scale.

Q &
A